- # Download and install Python from python.org

[Python.org Downloads](#)

Windows installation steps

1. Open the downloads page above.
2. Click **Download Python 3.x.x**.
3. Run the installer.
4. **Important:** check **“Add Python to PATH”** before clicking Install.
5. Click **Install Now**.
6. After installation, open Command Prompt and verify:

python –version

macOS installation steps

1. Download the macOS installer from the same page.
2. Open the .pkg file and follow the installer.
3. Verify in Terminal:

python3 --version

Linux

Many Linux distributions already include Python. To install/update:

- Ubuntu/Debian:

sudo apt update
sudo apt install python3

- Fedora:

sudo dnf install python3

Recommended next step

Install pip packages and create virtual environments:

python -m pip install --upgrade pip
python -m venv myenv

Official documentation:

Python Documentation

- ## Set up VS Code with Python extension

Install VS Code

Download and install the official editor:

Visual Studio Code

Choose the installer for your operating system and complete the setup.

Install the Python extension

Open VS Code, then install the official Python extension from Microsoft:

Python Extension for VS Code

You can also install it inside VS Code:

1. Open VS Code
2. Press Ctrl+Shift+X
3. Search for **Python**
4. Install the extension published by Microsoft

Verify Python is detected

1. Open VS Code
2. Press Ctrl+Shift+P

3. Type:

Python: Select Interpreter

4. Choose the Python version you installed from python.org

If Python does not appear, restart VS Code.

Create and run a Python file

1. Create a new file named:

hello.py

2. Add:

print("Hello, Python!")

3. Run it:
   - Click the ▶ Run button in the top-right
   - Or open Terminal in VS Code:

python hello.py

Optional but useful extensions

- Pylance Extension — better autocomplete and type checking
- Jupyter Extension — notebooks inside VS Code

Official setup guide

VS Code Python Tutorial

- **Write and run your first 'Hello World' program**

Install VS Code

Download and install the official editor:

[Visual Studio Code]

Choose the installer for your operating system and complete the setup.

Install the Python extension

Open VS Code, then install the official Python extension from Microsoft:

[Python Extension for VS Code]

You can also install it inside VS Code:

1. Open VS Code
2. Press Ctrl+Shift+X
3. Search for Python
4. Install the extension published by Microsoft

Verify Python is detected

1. Open VS Code
2. Press Ctrl+Shift+P
3. Type:

   Python: Select Interpreter

4. Choose the Python version you installed from python.org

   If Python does not appear, restart VS Code.

Create and run a Python file

1. Create a new file named:

   hello.py

2. Add:

   print("Hello, Python!")

3. Run it:
   - Click the ▶ Run button in the top-right
   - Or open Terminal in VS Code:

python hello.py

Optional but useful extensions

- Pylance Extension — better autocomplete and type checking
- Jupyter Extension — notebooks inside VS Code

Official setup guide

VS Code Python Tutorial

We use cookies

Cookies help this site function, measure usage, and support marketing. Manage your cookie preferences anytime. Learn more about our cookie policy.

- ## Create variables to store your personal information

```
# Personal information variables

name = "Rahul Sharma"
age = 22
city = "Delhi"
email = "rahul@example.com"
phone = "9876543210"

# Display the information
print("Name:", name)
print("Age:", age)
print("City:", city)
print("Email:", email)
```

```
print("Phone:", phone)
```

- **Build a simple calculator that adds two numbers**

```
# Simple Calculator for Addition

num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

sum = num1 + num2

print("The sum is:", sum)
```

- **Create a program that asks for your name and says hello back**

```
# Program to ask for name and greet the user

name = input("Enter your name: ")

print("Hello,", name + "!")
```

- **Practice fixing syntax errors in provided code examples**

1. Missing Colon :

Wrong Code

```
if age > 18
    print("Adult")
```

Correct Code

```
if age > 18:
    print("Adult")
```

---

2. Missing Closing Quote

Wrong Code

```
name = "Rahul
print(name)
```

Correct Code

```
name = "Rahul"
print(name)
```

---

3. Incorrect Indentation

Wrong Code

```
if True:
print("Hello")
```

Correct Code

```
if True:
    print("Hello")
```

---

4. Missing Parentheses

Wrong Code

```
print "Welcome"
```

Correct Code

```
print("Welcome")
```

---

5. Using = Instead of ==

Wrong Code

```
if number = 10:
    print("Number is 10")
```

Correct Code

```
if number == 10:
    print("Number is 10")
```

---

6. Unclosed Bracket

Wrong Code

```
numbers = [1, 2, 3
print(numbers)
```

Correct Code

```
numbers = [1, 2, 3]
print(numbers)
```

---

7. Misspelled Keyword

Wrong Code

```
pritn("Hello")
```

Correct Code

```
print("Hello")
```

---

8. Extra Indentation

Wrong Code

```
print("Start")
    print("Error")
```

Correct Code

```
print("Start")
print("No Error")
```